Essentials of

economics

Krugman • Wells • Olney

chapter 5

Elasticity

Prepared by:

Fernando & Yvonn Quijano

chapter 5

DRIVE WE MUST

Gassing up: A hard habit to break.

What you will learn in this chapter:

- The definition of **elasticity**, a measure of responsiveness to changes in prices or incomes
- The importance of the **price elasticity of demand**, which measures the responsiveness of the quantity demanded to price
- The meaning and importance of the **income elasticity of demand**, a measure of the responsiveness of demand to income
- The significance of the **price elasticity of supply**, which measures the responsiveness of the quantity supplied to price
- What factors influence the size of these various elasticities
- How elasticity affects the incidence of a tax, the measure of who bears its burden

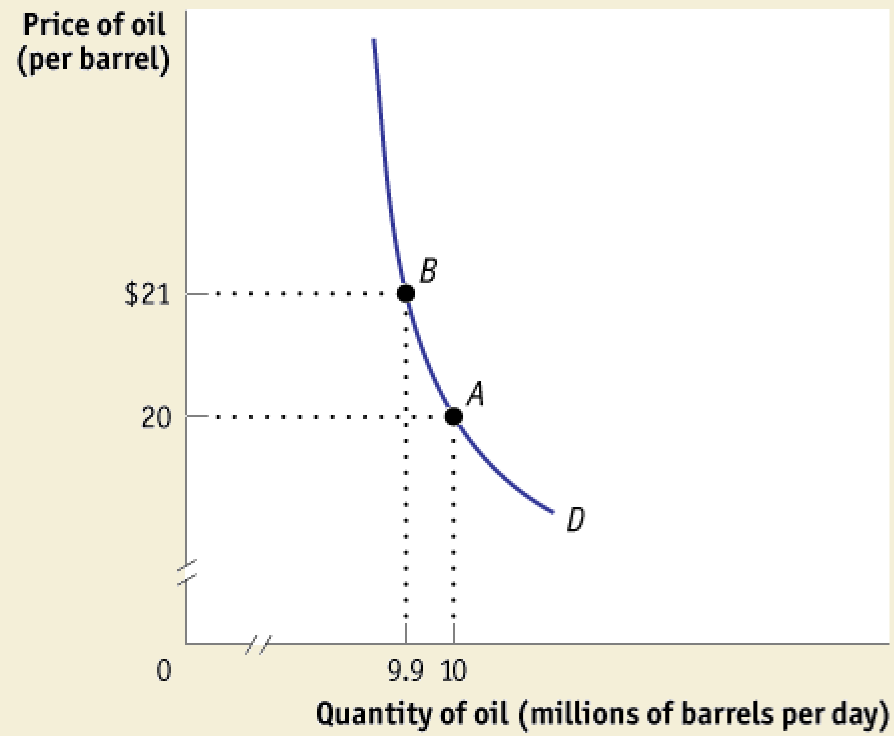
Defining and Measuring Elasticity

The Price Elasticity of Demand

Figure 5-1

The World Demand for Oil

At a price of $20 per barrel, the world quantity of oil demanded is 10 million barrels per day (point *A*). When price rises to $21 per barrel, world demand falls to 9.9 million barrels per day (point *B*).



Defining and Measuring Elasticity

The Price Elasticity of Demand

(5-1) $$\text{\% change in quantity demanded} = \frac{\text{Change in quantity demanded}}{\text{Initial quantity demanded}} \times 100$$

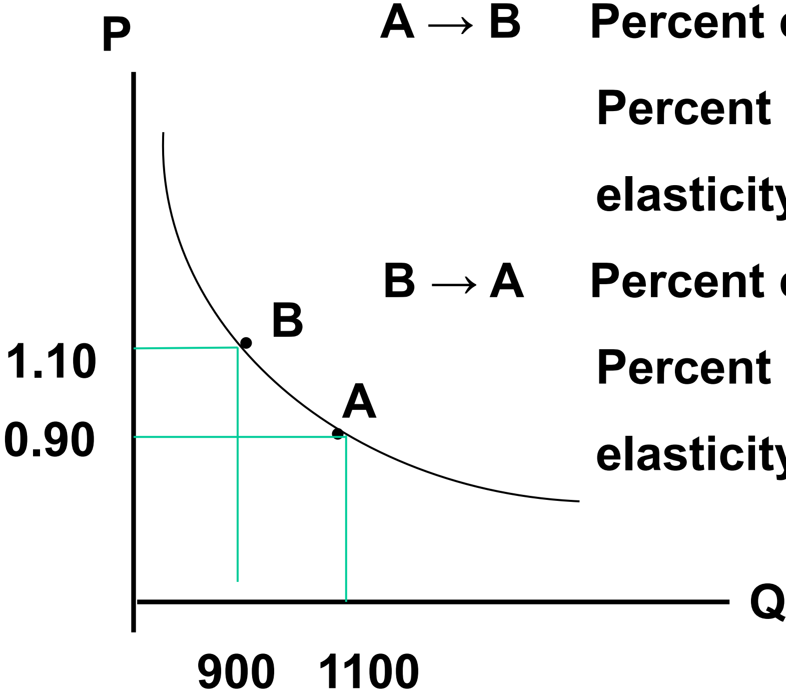
(5-2) $$\text{\% change in price} = \frac{\text{Change in price}}{\text{Initial price}} \times 100$$

(5-3) $$\textbf{Price elasticity of demand} = \frac{\text{\% change in quantity demanded}}{\text{\%change in price}}$$

The **price elasticity of demand** is the ratio of the percent change in the quantity demanded to the percent change in the price as we move along the demand curve.

Some Problems in Calculating Elasticities

	Price	Quantity
Situation A	$0.90	1,100
Situation B	$1.10	900



A → B Percent change in price: 0.2/0.9 * 100 %

Percent change in quantity: 200/1100 * 100%

elasticity = 0.82 (= 18.1 / 22.22)

B → A Percent change in price: 0.2/1.10 * 100 %

Percent change in quantity: 200/900 * 100%

elasticity = 1.22 (= 22.2 / 1.2)

Defining and Measuring Elasticity

Using the Midpoint Method to Calculate Elasticities

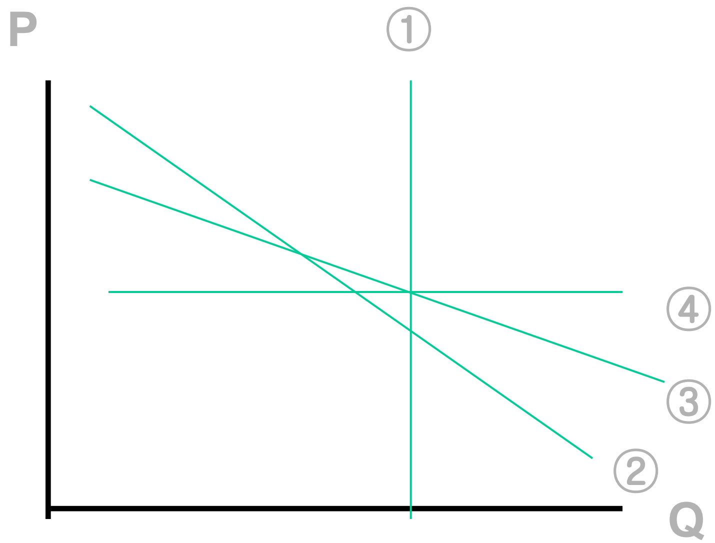
The **midpoint method** is a technique for calculating the percent change. In this approach, we calculate changes in a variable compared with the average, or midpoint, of the starting and final values.

(5-4) $$\%\text{ change in } X = \frac{\text{Change in } X}{\text{Average value of } X} \times 100$$

(5-5) $$\text{Price elasticity of demand} = \frac{\dfrac{Q_2 - Q_1}{(Q_1 + Q_2)/2}}{\dfrac{P_2 - P_1}{(P_1 + P_2)/2}}$$

Do the questions 1, and 2 on page 147.

Elastic and Inelastic Demand Curve



The Change of Q in response to the change of P

① : No elastic (perfectly no elastic)

② : Elastic

③ : More elastic

④ : Infinitely elastic (perfectly elastic)

economics in action

Estimating Elasticities

TABLE **5-1**

Some Estimated Price Elasticities of Demand

Good	**Price elasticity of demand**
Inelastic demand	
Eggs	0.1
Beef	0.4
Stationery	0.5
Gasoline	0.5
Elastic demand	
Housing	1.2
Restaurant meals	2.3
Airline travel	2.4
Foreign travel	4.1

Interpreting the Price Elasticity of Demand

What Factors Determine the Price Elasticity of Demand?

Whether Close Substitutes Are Available

Whether the Good Is a Necessity or a Luxury

Time

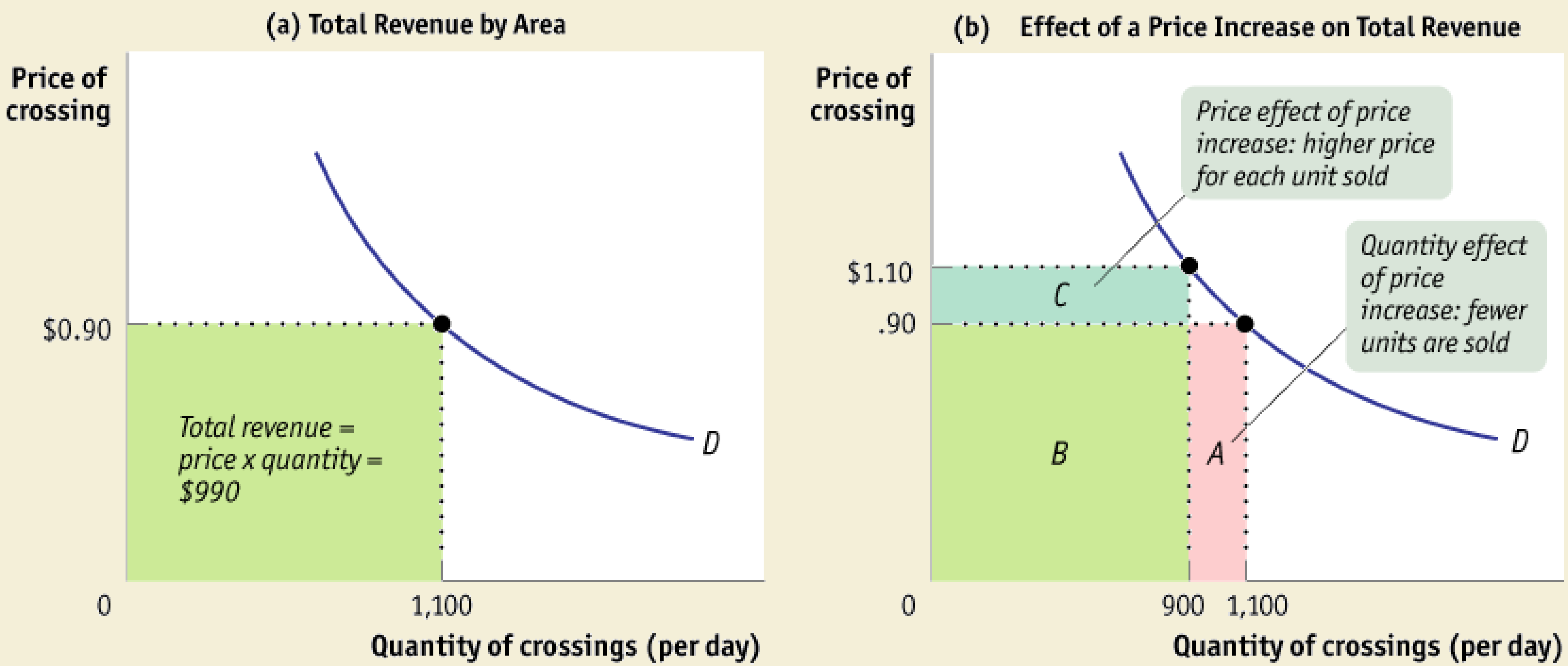
Proportion in consumer’s budget

5 chapter

Total Revenue and Price Elasticity of Demand

Total Revenue (R) = Price * Quantity Sold (Q)

Figure 5-4 Total Revenue



The green rectangle in panel (a) represents total revenue generated from 1,100 drivers who each pay a toll of $0.90. Panel (b) shows how total revenue is affected when the price increases from $0.90 to $1.10. Due to the quantity effect, total revenue falls by area *A*. Due to the price effect, total revenue increases by the area *C*. The overall effect can go either way, depending upon the price elasticity of demand.

Mathematics for Revenue and Elasticity

Revenue = Price * Quantity : R = P * Q

So, R is a function of P, but Q is also a function of P. Because when P increases, Q (=demand) decreases.

R = P * Q(P),

$$\frac{dR}{dP} = Q + \frac{dQ}{dP}P \quad , \; where \; \frac{dQ}{dP} < 0$$

If we divide this equation by Q,

$$\frac{dR}{dP}\frac{1}{Q} = 1 + \frac{dQ}{dP}\frac{P}{Q} \qquad Since \; elasticity \; e = \left|\frac{dQ}{Q} / \frac{dP}{P}\right| = \left|\frac{dQ}{dP} / \frac{P}{Q}\right|,$$

$$\frac{dR}{dP}\frac{1}{Q} = 1 - e \quad , \; (remember \; that \; \frac{dQ}{dP} < 0\,)$$

$$\frac{dR}{dP} = Q\,(1 - e\,)$$

The Interpretation of the $\frac{dR}{dP} = Q(1-e)$

(1) $\frac{dR}{dP} > 0$ means that, as price goes up, the total revenue increases (even though the demand Q decreases).
So, when the price elasticity of demand is inelastic ($e < 1$),
The seller's revenue will increase if he raises the price.

(2) $\frac{dR}{dP} = 0$ means that although the price changes, the total revenue does not change. This is when the elasticity is 1.
Also when the elasticity is 1, the total revenue becomes maximum or minimum.

(3) $\frac{dR}{dP} < 0$ means that as price goes up, the total revenue decreases.
So if the elasticity is elastic ($e > 1$) , the seller's revenue will decrease as he increases price. (Even though the price increases, its response, i.e., demand decreases too much.)

What Factors Determine the Price Elasticity of Demand?

(1) Are there close substitutes?

(2) Is the good a necessity or a luxury?

(3) Is the share of the income spent on the good high?

(4) Time

Other Demand Elasticities

The Cross-Price Elasticity of Demand

The **cross-price elasticity of demand** between two goods measures the effect of the change in one good's price on the quantity demanded of the other good. It is equal to the percent change in the quantity demanded of one good divided by the percent change in the other good's price.

(5-7) $$\text{Cross-price elasticity of demand between goods A and B} = \frac{\%\text{ change in quantity of A demanded}}{\%\text{ change in price of B}}$$

It is important to remember that the Cross-price elasticity can be either positive or negative:

e_{yx} (X Price elasticity on demand for Y) > 0 : Substitutes

e_{yx} (X Price elasticity on demand for Y) < 0 : Complements chapter 5

Other Demand Elasticities

The Income Elasticity of Demand

The **income elasticity of demand** is the percent change in the quantity of a good demanded when a consumer's income changes divided by the percent change in the consumer's income.

(5-8) $$\text{Income elasticity of demand} = \frac{\%\ \text{change in quantity demanded}}{\%\ \text{change in income}}$$

The demand for a good is **income-elastic** if the income elasticity of demand for that good is greater than 1.

The demand for a good is **income-inelastic** if the income elasticity of demand for that good is positive but less than 1.

If income elasticity is negative, it means the good is a inferior good. A normal good will have a positive income elasticity.

The Price Elasticity of Supply

Measuring the Price Elasticity of Supply

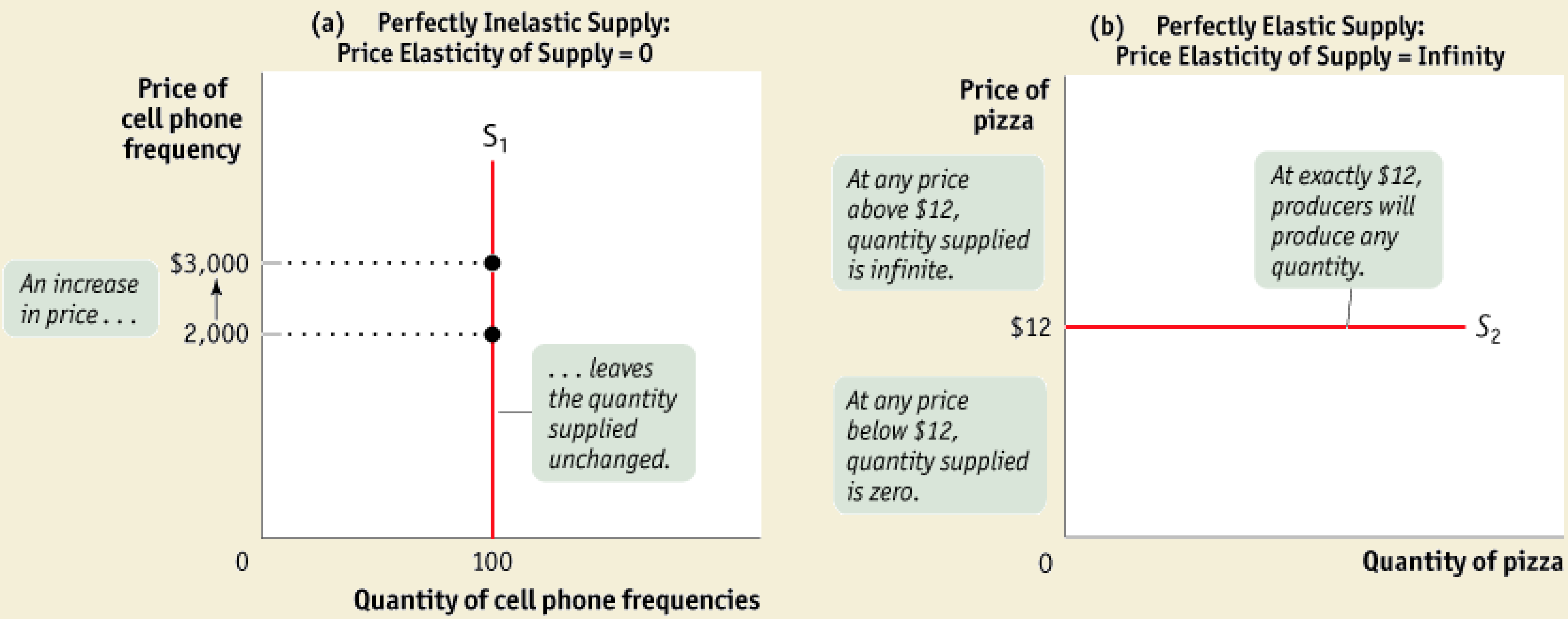
The **price elasticity of supply** is a measure of the responsiveness of the quantity of a good supplied to the price of that good. It is the ratio of the percent change in the quantity supplied to the percent change in the price as we move along the supply curve.

(5-9) $$\text{Price elasticity of supply} = \frac{\%\ \text{change in quantity supplied}}{\%\ \text{change in price}}$$

The Price Elasticity of Supply

Measuring the Price Elasticity of Supply

Figure 5-5 Two Extreme Cases of Price Elasticity of Supply



Panel (a) shows a perfectly inelastic supply curve, which is a vertical line. The price elasticity of supply is zero: the quantity supplied is always the same, regardless of price. Panel (b) shows a perfectly elastic supply curve, which is a horizontal line. At a price of $12, producers will supply any quantity, but will supply none at a price below $12. If price rises above $12, they will supply an extremely large quantity.

The Price Elasticity of Supply

Measuring the Price Elasticity of Supply

There is **perfectly inelastic supply** when the price elasticity of supply is zero, so that changes in the price of the good have no effect on the quantity supplied. A perfectly inelastic supply curve is a vertical line.

There is **perfectly elastic supply** when even a tiny increase or reduction in the price will lead to very large changes in the quantity supplied, so that the price elasticity of supply is infinite. A perfectly elastic supply curve is a horizontal line.

What Factors Determine the Price Elasticity of Supply?

The Availability of Inputs

Time

An Elasticity Menagerie (Table 5-3)

Name	Possible values	Significance
$\text{Price elasticity of demand} = \dfrac{\%\text{ change in quantity demanded}}{\%\text{ change in price}}$ (use absolute value)		
Perfectly inelastic demand	0	Price has no effect on quantity demanded (vertical demand curve).
Inelastic demand	Between 0 and 1	A rise (decrease) in price increases (decreases) total revenue.
Unit-elastic demand	Exactly 1	Changes in price have no effect on total revenue.
Elastic demand	Greater than 1, less than ∞	A rise (fall)in price reduces (increases) total revenue.
Perfectly elastic demand	∞	A rise in price causes quantity demanded to fall to 0. A fall in price leads to an infinite quantity demanded (horizontal demand curve).
$\text{Cross-price elasticity of demand} = \dfrac{\%\text{ change in quantity } \textit{of one good} \text{ demanded}}{\%\text{ change in price } \textit{of another good}}$		
Complements	Negative	Quantity demanded of one good falls when the price of another rises.
Substitutes	Positive	Quantity demanded of one good rises when the price of another rises.

An Elasticity Menagerie

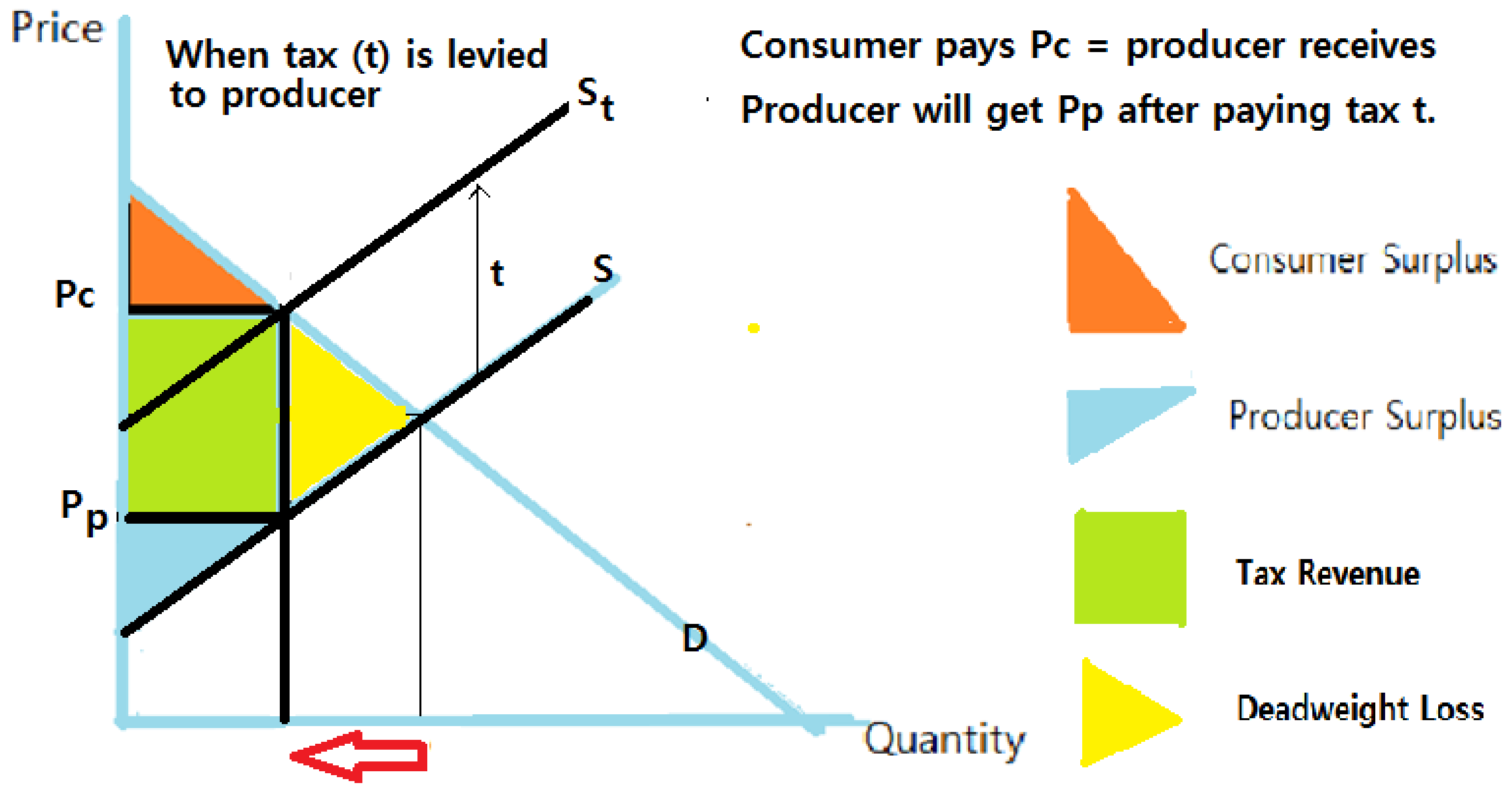
TABLE **5-3 cont'd.**

Name	Possible values	Significance
$$\textbf{Income elasticity of demand} = \frac{\textbf{\% change in quantity demanded}}{\textbf{\% change in income}}$$		
Inferior good	Negative	Quantity demanded falls when income rises.
Normal good, income-inelastic	Positive, less than 1	Quantity demanded rises when income rises, but not as rapidly as income.
Normal good, income-elastic	Greater than 1	Quantity demanded rises when income rises, and more rapidly than income.
$$\textbf{Price elasticity of supply} = \frac{\textbf{\% change in quantity supplied}}{\textbf{\% change in price}}$$		
Perfectly inelastic supply	0	Price has no effect on quantity supplied (vertical supply curve).
	Greater than 0, less than ∞	Ordinary upward-sloping supply curve.
Perfectly elastic supply	∞	Any fall in price causes quantity supplied to fall to 0. Any rise in price elicits an infinite quantity supplied (horizontal supply curve).

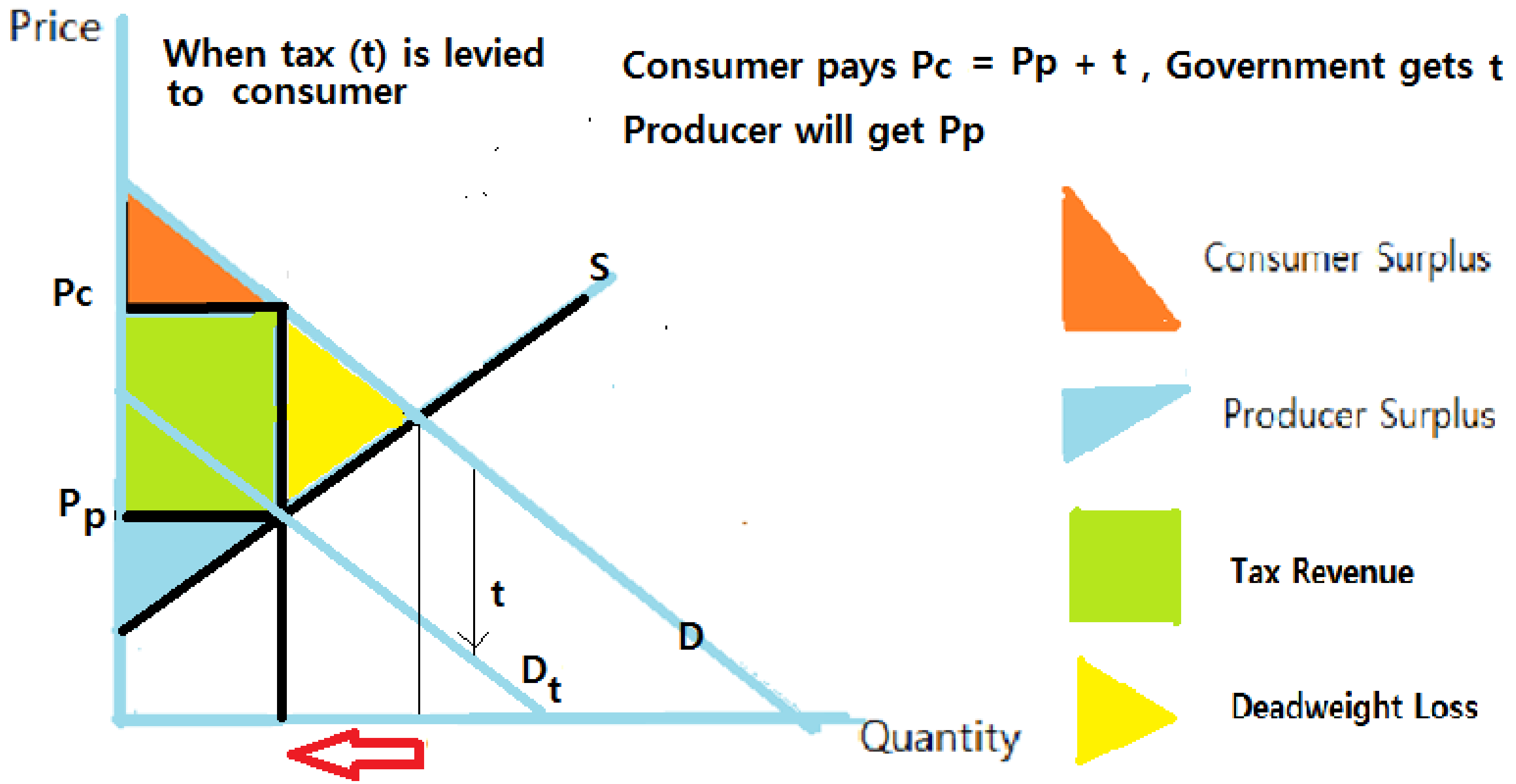
Excise Tax (1) Tax to Producer

Consumer pays Pc = producer receives

Producer will get Pp after paying tax t.



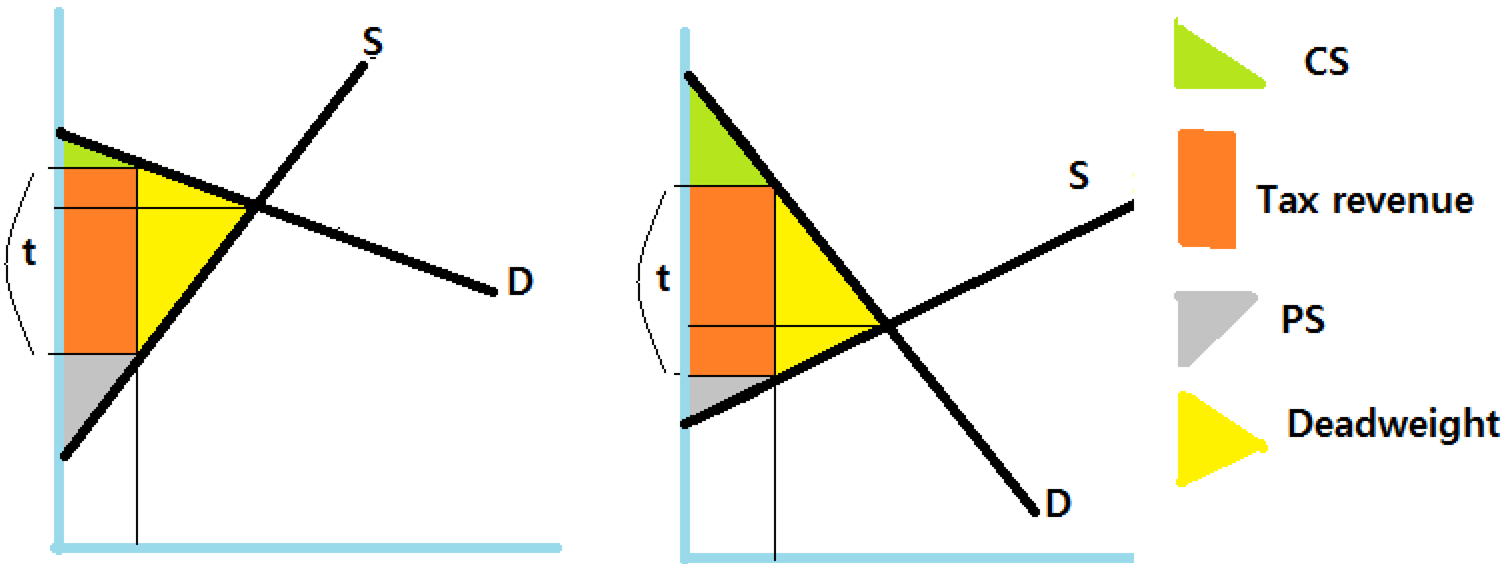
Excise Tax (2) Tax to Consumer



Tax Incidence : Who loses (or burdens) more ?

Demand elastic, Supply inelastic

Demand inelastic Supply elastic

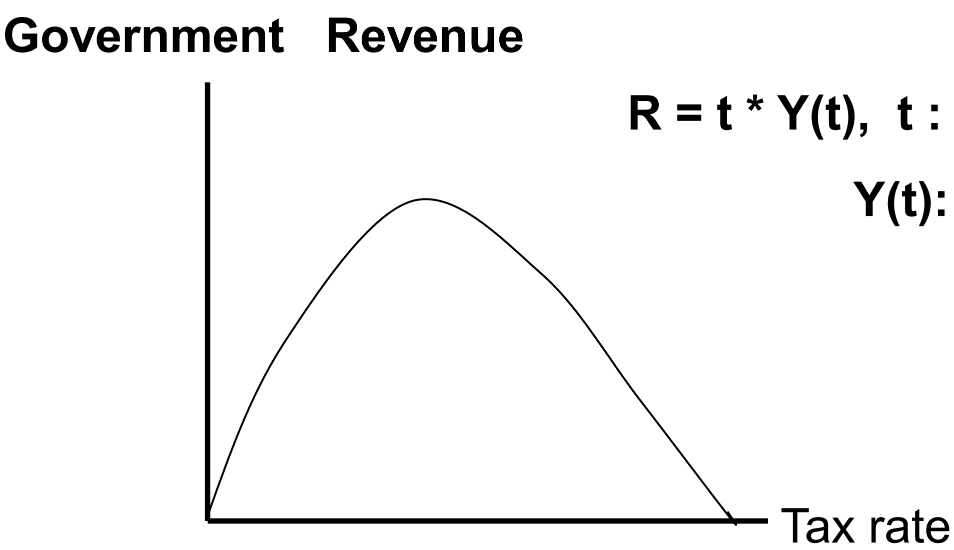


Tax burden goes mainly to producer

Tax burden goes mainly to consumer

Laffer Curve

Does a increase of tax rate increase tax revenue?



$R = t * Y(t)$, t : tax rate,

$Y(t)$: National Income

KEY TERMS

Price elasticity of demand
Midpoint method
Perfectly inelastic demand
Perfectly elastic demand
Elastic demand
Inelastic demand
Unit-elastic demand
Total revenue
Cross-price elasticity of demand
Income elasticity of demand
Income-elastic demand
Income-inelastic demand
Price elasticity of supply
Perfectly inelastic supply
Perfectly elastic supply